

Lectures

on

“Philosophical Propaedeutics”

from the Academic Year 1860–61

translated from the Danish by H. Halvorson

R. Nielsen

Copenhagen: Published by the Gyldendal Bookshop (F. Hegel), 1862

Contents

Translator's Note on Terminology	5
Preface	9
I. The Concept of Philosophy	11
A. The Doctrine of Contradiction	11
B. The Doctrine of Possibility	15
C. Dialectic.	21
Something and Other	22
Concept and Existence	27
II. Philosophy and the Particular Sciences	33
General Remarks	33
A. Philosophy's Relation to Mathematics: The Problem of Infinity	39
a) The Concept of Infinity	39
b) The Problem of Infinity	43
b). The Analysis of the Infinite	46
B. Philosophy's Relation to Chemistry: The Problem of Change	47
a) Chemistry as a Distinct Science	47
α) The Origin of Chemistry	47
β) Chemical Forces	47
γ . Chemical Combinations	47
b) The Problem of Change in Chemistry	47
α) Atom and Equivalent	47
β) Atomic Constitution and Crystal Form	47
γ) Atom and Substance	47
c) The Problem of Change in Philosophy	47
α) The Atom as Point of Discretion	47

β) The Atom as Point of Function	47
γ) The Atom as the Substantial Moment	47
Result	47
C. Philosophy's Relation to Physiology: The Problem of Life	47
a) The Essence of Life	47
α) The Organism	47
β. The Organic Type	47
γ) The Organic Purpose	47
b) The Stages of Life	47
α) Plant Life	47
β) Animal Life	47
γ. Human Life.	47
c) Soul and Body.	47
α) Metaphysical Theories.	47
β) Psychophysical Investigations.	47
γ) The Limits of Human Knowledge.	47
III. Philosophy and Academic Study	49
Scientific Thinking	49
The Connection of the Sciences	49
Scientific Cultivation	49
Contents (the book's own Indhold)	51

Translator's Note on Terminology

Nielsen lectured in 1860–61 in a philosophical Danish that had been formed on German models, and much of his vocabulary is a set of Danish calques on Kantian and Hegelian terms. A few of these words have modern English cognates that mean something quite different, and a reader who takes them at face value will be misled. The renderings below are used consistently throughout; a dagger footnote marks the first occurrence of each in the text.

***den sunde Forstand* — “sound understanding”**

Literally “the healthy understanding”; the standing Danish equivalent of German *der gesunde Menschenverstand*, and close to English “common sense.” But *Forstand* is *Verstand*, the understanding as opposed to reason, and Nielsen’s whole argument turns on its being a *faculty* that can be shown to contradict itself — not a stock of shared opinion. He almost always sets it in quotation marks, and those are preserved. The fuller form *den sunde Menneskeforstand* is given as “sound human understanding.”

***Anskuelse* — “intuition”**

Anschauung: the immediate presentation of an object to sense, in Kant’s technical sense. Never “opinion” or “view,” which is what the cognate suggests in modern Danish.

***Forestilling* — “representation”**

Vorstellung: the mental image or presentation, standing between sensation and concept. Nielsen’s argument at pp. 10–11 depends on the contrast between what can be *thought* and what can be *represented* — a round square can be thought, he says, but not represented.

***sandselig, Sandsning, Sandseindtryk* — “sensuous, sensation, sense-impression”**

sinnlich: pertaining to the senses. “Sensuous” here carries none of the modern connotation of “sensual.”

Skarpsindighed / Spidsfindighed – “acumen” / “over-subtlety”

A matched pair in Danish (*skarp* = sharp, *spids* = pointed) that English cannot match morphologically. The first is the virtue of exact analysis; the second is its vice — subtlety that dazzles by distinctions only half true. Distinguishing them is the business of Part I.A, and Nielsen insists the same utterance may be either.

Menen / Viden – “opining” / “knowing”

The Platonic $\delta\acute{o}\xi\alpha/\acute{\epsilon}\pi\iota\sigma\tau\acute{\eta}\mu\eta$ distinction. Nielsen puns on it at p. 22 (“the question is not of opining but of knowing — still, what is your opinion?”).

Ophæve, Ophævelse – “sublate, sublation”

aufheben: at once to cancel, to preserve and to raise up. Where the context is plainly non-technical the ordinary “cancel” or “annul” is used instead.

Tvetydig / Tvesidig – “ambiguous” / “two-sided”

Another pair the English breaks: literally “two-meaning” and “two-sided.” Nielsen’s definition of dialectic at p. 15 is that it is the science which discovers the *two-sided* in the *two-meaning*.

Grund / Følge – “ground” / “consequent”

Grund/Folge, the logical pair, not cause and effect.

dialektisere – “dialecticize”

Nielsen’s own verb, and deliberately ungainly in Danish too: to derive one thing from another by dialectic. It is kept rather than paraphrased, since the dialogue of pp. 24–28 turns on the question of what can and cannot be so derived.

Grundsætning / Sætning – “fundamental principle” / “proposition”

Sætning does duty for both “sentence,” “proposition” and “theorem”; where Nielsen means one of the laws of thought, “principle” is used.

diversi respectus

Left in Latin, as Nielsen leaves it: the scholastic maxim *diversi respectus tollunt omnem contradictionem* — different respects remove every contradiction. It is the hinge of the first dialogue, and translating it would blunt the joke that B. has never troubled to study what a *respectus* is.

Nielsen cites three of his own earlier books by short title, without further identification: *Propæd.* or *Propædeut.* is *Philosophisk Propædeutik i Grundtræk* (Copenhagen 1857); *Philos. og Mathem.* is *Philosophie og Mathematik*; *Mathem. og Dialekt.* is *Mathematik og*

Dialektik. These are left in Danish, as he gives them. All page references in such citations are to the Danish originals.

Nielsen's own footnotes are marked, as in the print, *) and **); translator's notes are marked with a dagger and open with the words *Translator's note*.

Preface

Should the present work attract any attention, the interest taken in it will doubtless be of two kinds. With the question of the possible worth of its contents, most readers, keeping to the letter of the title, will surely join a second and more practical question: *on what principles, and to what extent, was the philosophical propaedeutic here delivered to the students?*

As regards this latter question, it will not be enough to remark merely that the motives, leading ideas and whole design of the present exposition correspond to the oral lectures, but that in matters of detail a good deal has been altered, something added, something else omitted, and so forth; the reader will think himself entitled to a clearer account.

It is therefore expressly noted that, apart from a few less considerable editorial changes, the whole of the first part, pp. 1–30, is a straightforward reproduction of the lectures as delivered. The same holds of the introductory piece of the second part, *General Remarks*, pp. 30–39; of Philosophy's Relation to Mathematics, pp. 39–63; of Philosophy's Relation to Chemistry, pp. 81–100 and pp. 135–177; and of Philosophy's Relation to Physiology, pp. 177–180, pp. 199–288, pp. 386–437, pp. 475–482.

In the presentation of the remaining portions, by contrast, considerable changes have been made. The section on the psychophysical investigations was not touched upon at all in the oral lectures; sharper and further-reaching analyses have been introduced here and there; and it goes without saying that by far the greater part of the contents of the notes was added in the course of the revision.

Since the effort to give the treatment of the physiological problems in particular that fullness which investigations of so intricate a nature can hardly do without brought with it a considerable expansion of the second part, the third part — Philosophy and Academic Study — has been confined to a bare statement of its moments, and is appended as a separate part only in order to exhibit the course of the whole.

11 June 1862.

R. Nielsen.

I.

The Concept of Philosophy.

A.

The Doctrine of Contradiction.

What is philosophy? “A science full of contradictions!” However much this definition may seem, viewed from one side, designed to pass sentence on philosophy altogether, it is nonetheless well founded, both in the history of the science and in the nature of the case. For any other science such an admission would be annihilating; for philosophy it furnishes an essential starting point, a dependable fulcrum, a firm and unshakable foundation. That philosophy must contain contradictions is a matter of course, since it is precisely the science which everywhere discovers and exhibits contradictions — in existence itself no less than in sound understanding’s[†] grasp of existence (Propæd. pp. 40–44)[†] — and it is just in this respect that philosophy is to be understood as a *general doctrine of contradiction*.

As a doctrine of contradiction, however, philosophy must not merely discover and exhibit contradictions; it must also test them, for there are differences among contradictions. Some are superficial, unnecessary, easily avoided; others are insistent and not to be waved aside. Some creep in through thoughtlessness, others are manufactured by over-subtlety, others again brought to light by acumen.[†] Some contradictions play about

[†] *Translator’s note.* *Den sunde Forstand*: the Danish equivalent of German *der gesunde Menschenverstand*, and close to English “common sense.” But *Forstand* is *Verstand*, the understanding as against reason, and Nielsen’s argument turns on its being a faculty that can be shown to contradict itself. He sets it in quotation marks almost everywhere, and those are kept. See the Translator’s Note on Terminology.

[†] *Translator’s note.* Nielsen’s short titles for his own earlier books, cited throughout without further identification: *Propæd.* (also *Propædeut.*) is his *Philosophisk Propædeutik i Grundtræk* (Copenhagen 1857); *Philos. og Mathem.* is *Philosophie og Mathematik*; *Mathem. og Dialekt.* is *Mathematik og Dialektik*. Page numbers refer to the Danish originals.

[†] *Translator’s note.* *Spidsfindighed* and *Skarpsindighed* form a matched pair in Danish (*spids* = pointed, *skarp* = sharp) that English cannot match. The first names the vice of subtlety that dazzles by half-true

as in a game of thought; others mark essential knots, difficult problems not easily untied.

The following examples may serve to show how necessary it is, where contradictions are concerned, to distinguish the superficial from the essential, the acute from the over-subtle.

The Liar (ψευδόμενος). This contradiction is put as follows: when someone says that he is lying, is he lying in that very statement, or is he telling the truth? The answer demanded is either yes or no. "Yes, he is lying!" — And he says so himself, calls the thing by its right name, says what is true, and hence is not lying. "No, he is not lying, he is speaking the truth!" — Then one must believe what he says; but what he says is precisely that he is lying; and if it is true that he is lying, then he is not speaking the truth.

That there is a contradiction here, sound understanding can readily grasp; but what sort of contradiction is it, and where does the fault lie?

Equal Magnitudes. That a magnitude is equal to itself, and that two, three .. magnitudes are equal to one another — in this sound understanding sees no contradiction. But if it were asserted that two magnitudes cannot possibly be equal to one and the same third, and hence equal to each other, because that would be a contradiction, sound understanding would feel itself affronted. Is there then really any contradiction whatever in the proposition just cited? To see what is actually being spoken of here, one must expressly note that the talk is not of things that have magnitude but of magnitudes. If the talk were not of magnitudes themselves but of things that have magnitude — an apple, say, a triangle, an hour, and so on — there would certainly not be a trace of contradiction. That two triangles, each of which is equal to one and the same third, are equal to one another, goes without saying. So long as we have to do with concrete magnitudes, difference and plurality can coexist perfectly well with equality; but when it is pure magnitudes that are in question, the matter stands otherwise. A plurality of magnitudes necessarily presupposes that the one can be distinguished from the other: without mutual difference, no plurality. But by what should pure magnitudes be distinguished from one another if not by size? If the magnitudes are equal to one another, then every glimmer of difference has vanished, and with it the plurality: to say that several magnitudes are equal to one another is therefore to say, in other words, that they are not several but one and the same magnitude, which, however many times it be repeated, is not thereby multiplied but remains the very same.

Should "sound understanding" wish to reject this consequence because there is something offensive about it, that can hardly be done out of hand by a mere dictum of

distinctions; the second the virtue of exact analysis. Distinguishing them is the business of this section, and Nielsen insists that one and the same utterance may be either.

authority; a sensible rejection must set about showing wherein the difficulty consists and how it is to be removed. If one and the same magnitude can, by being named twenty times, become twenty magnitudes equal to one another, then one and the same species must likewise be able, by being named twenty times, to become twenty species; but if it is not enough to repeat the same magnitude — if in repeating it a difference must also be thought — then where is that difference to come from?

The Instant.[†] If the old question — when did Socrates die? — is answered by pointing to the year of his death, sound understanding will hardly suspect any contradiction in that. Unless indeed it be that Socrates has no year of death at all, inasmuch as he did not take a year over dying: he was hale and healthy through the whole of that last year, until he drank from the cup. But if, in accordance with a well-known and warranted usage, one means by his year of death the year in which the day of his death fell, then there is no contradiction here. If it be objected that he did not take a whole day over dying either — well then, let one think of the hour of his death; and if one will go to the very limit, of the instant of his death. So then: when did Socrates die? “At the instant he ceased to live!” But this answer, which tells us only that he died when he died, is no answer at all. The question concerns the point of time in Socrates’ existence at which his death occurred. To specify this point of time, four cases may be conceived: either he died α) while he was still living, or β) when he was already dead, or γ) when he was partly living and partly dead, or δ) when he was neither living nor dead. But α) to die while one is still living, to be a living dead man, is surely a contradiction. And to die when β) one is already dead is surely a contradiction too. The instant at which Socrates died must therefore have been the instant when, lying on the border between life and death, he was in such a condition that one part of him — his feet, say — was already dead while another part was still alive. By this explanation, however, the question is not answered but only pushed further back; for the question then is: when did the already dead part die? If the answer is that the already dead part died when it was still partly alive and partly dead, then the contradiction concealed in case γ) becomes obvious. And to conceive Socrates in a condition in which he was neither living nor dead is, finally, likewise to conceive a contradiction (δ); for how long does a living thing live? Until it dies!

“A philosophy that stands in open contradiction with ‘sound understanding’ is false; a philosophy that conforms itself immediately to sound understanding is superficial. Philosophy does not wish arbitrarily to contradict the natural understanding, but only to show

[†] Translator’s note. *Øieblikket*: literally “the eye-blink,” the smallest unit of time. “Instant” is used rather than “moment” throughout, to keep the word clear of the Hegelian *Moment* that Nielsen also employs.

that so-called sound sense, without having any inkling of it, incessantly contradicts itself.” (Propæd. p. 41).

In so far as contradictions of these various kinds develop in the course of thinking, and their origin is viewed from the subjective side, the psychological share that reflection has in the contradictions offensive to “sound sense” must be determined more closely. The aberrations belonging here may be briefly characterized as the *eristic*, the *over-subtle* and the *sophistical*.

The striking mark of eristic is a one-sided bent for quarrel and strife, a delight in shining everywhere by objections, in raising difficulties, in tearing down what is generally accepted and stubbornly defending one’s own divergent claims. As for ancient philosophy, it is above all the Megarian school, akin to the Eleatic, that has earned the name of eristic; but neither can it be denied that every distinctively developed form of philosophy has its polemical element, and that eristic, for all its dark sides, is to that extent of unmistakable importance in the development of the science. For insight and knowledge it is not enough that one form what is commonly called a reasonable opinion about things, and, by combining sense-impressions with natural judgment, be led to distinguish correct from incorrect, acceptable from objectionable opinions; by means of opinions one attains only the probable, not the true. Whoever would rise from mere opining to knowing[†] must arm himself against the deceptions of sense, let thought test what has been accepted, and acknowledge as true only what admits of being thought — declaring untrue, without regard to the multitude’s otherwise unimpeachable opinions, everything that does not admit of being thought. That philosophy, on these terms, must come into manifold conflict with sound human understanding is evident. (Cf. Philos. og Mathem. p. 12, note; Mathem. og Dialektik pp. 34–35, pp. 116–117, pp. 131–32, etc.) Rigorous analysis of concepts demands acumen; but acumen that is one-sided and unfruitful, dazzling by distinctions which are subtle yet only half true or even untrue, yields over-subtlety. Easy as it is to state the difference between acumen and over-subtlety so long as one keeps to generalities, it is just as impossible to lay down a rule valid for every particular case; for one and the same utterance may, seen from one side, look over-subtle, and yet, seen from another, bear the stamp of sound and genuine acumen. The contradiction set up in “the Liar” may serve as an example. If the task is to drive home the truth that one should not undertake off-hand to answer every question, since there are questions whose apparently clear and easily grasped sense proves on closer inspection to be senseless, then “the Liar” may stand as a model of

[†] Translator’s note. Menen and Viden, the Platonic pair δόξα/ἐπιστήμη. Nielsen returns to the pun at p. 22.

genuine acumen. If, on the other hand, one guilelessly takes up the answer demanded — and does so in such a way that, watching with astonishment as the affirmation turns into a denial and the denial back into an affirmation, one ends by imagining oneself before some deep riddle, some exceedingly intricate knot whose solution may perhaps shed a hitherto unknown light upon “the eternal truths” — then over-subtlety has outwitted sound understanding and caught it in its web.

In secret league with over-subtlety and hair-splitting we find sophistry. The sophist is not content to exhibit contradictions; he manufactures contradictions, plays with contradictions, draws inferences, and dazzles by false inferences. For it is one thing to bring contradictions into relief, another to misuse them in the service of sophistic; how great the difference is may be seen, for instance, from what was said above about “the Instant.” No sensible person can regard it as sophistry to call attention to the contradictions upon which thought must necessarily strike when it tries to give an account of the meaning of the instant; only when it is a question of the use made of the contradictions does sophistry become recognizable, by what is false, untenable and dazzling in the inferences. Thus if anyone were to infer, from the contradictions contained in the thought of the instant of death, the unthinkability of death, and from the unthinkability of death its impossibility, and were then to build upon that a new proof of the immortality of the soul, fitted out with a peculiar acumen, we should have to consider him a sophist.

B.

The Doctrine of Possibility.

If philosophy as a doctrine of contradiction is not to dissolve into over-subtlety and sophistry, it must always find itself able, from one side or another, to master the contradictions. It must then be able to determine their relation both to things and to our reflections upon things, both to objective being and to subjective thinking; in seeing through the essence of contradiction it must discover the marks of what is in itself thinkable and what is in itself unthinkable, and thereby, so to speak, wring from contradiction a dependable measure of the acceptable and the objectionable, the reasonable and the absurd, the possible and the impossible: a thoroughgoing doctrine of contradiction is inseparable from a *general doctrine of possibility*.

What, then, is philosophy? “A doctrine of the possible.” But since philosophy cannot possibly be a doctrine of everything possible, this Wolffian definition is satisfactory

only on condition that the concept of possibility — its extension, its content, and its relation to contradiction — be exactly determined. The principle that everything is possible which does not contain a contradiction (*possibile est, quod non implicat contradictionem*) — which in turn presupposes that everything containing a contradiction is impossible — will, on the strength of what has gone before, hardly furnish a dependable guiding line, whether dogmatic or critical.

In the first place it is self-evident that the principle as stated specifies possibility only formally, and is therefore unfit to orient us with respect to the real possibilities. Even if for the time being we declare everything that contains a contradiction impossible, and everything that does not contain a contradiction possible, the concept of possibility so given will still be so negative, so empty and unsettled, that it cannot be brought into any positive connection whatever with the actual. To determine the possibilities that hold in actuality, it is necessary to know actual things together with their conditions and laws; and knowledge of actual things is reached only by the road of experience. Without the correctives of experience, dogmatizing thought with its empty concepts will be as readily tempted to set the real possibilities too narrow bounds as to allow them too wide a play. Or who, back in the days when Wolff was fixing the bounds and defining the possibilities, would have seen the possibility that ships should be able to go both against wind and against current; that carriages without horses should be able to run — not like horses, but swift as flying birds; or the possibility that a man in London should be able to correspond with a man in Calcutta as quickly and easily as if they were talking together? So long as philosophy, with its formal concepts and definitions, attempts to exhaust existence and to set itself up as judge over what is possible and impossible in actuality, so long will sound human understanding have manifold occasion for the triumphant application of Hamlet's words: *There are more things in heaven and earth, Horatio, than are dreamt of in your philosophy.*

But even if philosophy gives up trying to fathom the positive, empirical content of the real possibilities, it may still contribute essentially to purifying and heightening our knowledge of the possible, provided only that it can secure for us the negative side of possibility by determining the impossible. Now if the impossible can be known only as what is in itself unthinkable, and if the unthinkable must always be recognizable by contradiction, then the fundamental question presses itself upon us of its own accord: *how, then, does the contradictory stand to the thinkable?* Here we are at a decisive turning point.

In presupposing without more ado the unity of thinking and being, dogmatic phi-

losophy proceeds at the same stroke from the assumption that the contradictory is unconditionally an unthinkable, and the unthinkable an impossible. But is the contradictory really unthinkable in every respect?

“What does contradiction mean? An existing nothing is a contradiction; a round square is a contradiction; a flying horse is a contradiction too. And yet these contradictions are of different kinds.” (Propæd. p. 43).

If what is contradictory in itself is straightway unthinkable, how then does thought come to apprehend this unthinkable? How does it happen that we not only understand what is meant by the word “contradiction,” but also see clearly what lies in the essence of contradiction? If we set, say, “something” against “nothing,” “that which is” against “that which is not,” thought finds it just as easy to grasp the negative as to grasp the positive; and if we set the proposition “that which is, is” against the proposition “that which is, is not,” one notices at once that a conflict between subject and predicate is as evident to thought as an agreement. But if thought apprehends the conflict as readily as the agreement, sees through an existing nothing as clearly as an existing something, and so far finds the former as thinkable as the latter — how then can what is in itself conflicting and contradictory show itself to thought as an unthinkable, and this unthinkable coincide with the impossible?

To extricate oneself from the difficulty, let one try letting representation come to the support of thinking.[†] Pure, abstract thinking, detached from all intuition and representation, does indeed grasp the negative as readily as the positive, since the negative is precisely abstraction’s own abstract product; but when the light of intuition dawns, when representation joins in, the opposition between the positive and the negative, between something and nothing, becomes conspicuous. For something admits of being intuited, whereas nothing, pure nothing, does not; to the positive there answers a representation, but nothing — the negative — repels every representation, and is for that reason in turn repelled and rejected by representation.

How thinking, by being combined with representation, comes to apprehend the contradictory as the unthinkable, and the unthinkable as the impossible, may be seen from the example of the round square. The proposition “the square is round” can indeed be thought, in so far as the predicate’s conflict with the subject can be thought; but the round square cannot be represented, since the representation of the round displaces

[†] *Translator’s note.* *Foretelling* is *Vorstellung*, the mental presentation or image standing between sensation and concept; *Anskuelse*, just below, is *Anschauung*, intuition in Kant’s technical sense — never “opinion” or “view.” The whole argument of this page turns on the contrast between what can be *thought* and what can be *represented*.

and effaces the representation of the square. Thought, seeing that the round square makes a demand upon representation which representation must reject, now brands the contradiction as something objectionable, declares the round square an unthinkable, and this unthinkable an impossible. That it is precisely intuition and representation which in such cases must help thought to discover the unthinkable is seen not only from propositions in whose components representation manifestly rules — "black is not white," "thick is not thin" — but also from examples that do not directly display the contradiction in their words and manner of expression, as when one asks: is a regular hexahedron possible?[†] is it possible for a line parallel to an ellipse to be an ellipse? and so on.

If only we could rely on intuition and representation to furnish, always and in every case, infallible marks of the thinkable and the unthinkable, we should at least have something firm to hold to here; but now look at the example of the flying horse! Natural sense surely feels that under the name "flying horse" a contradiction lies hidden. But to whom is sound human understanding now to turn to have the contradiction cleared up? To thinking! That is no help: pure thinking thinks everything, the unthinkable included. To intuition! No help either: intuition lives with the poet, and the poet rides Pegasus, and Pegasus reconciles intuition with every winged horse. To a union of thinking and intuition! But how can a union of two unreliaables yield anything reliable? To sense-impressions! Sense-impressions can indeed, in union with thinking, teach us something about the possible, in so far as they teach us something about the actual, of course; but what do they teach us about the merely possible, and what could they possibly teach us about what is in itself contradictory and impossible?

To determine the relation of possibility to contradiction, it is above all necessary to grasp *the principle of contradiction* — its form, its validity, its application.

In formal logic we find the propositions of identity and of contradiction set out so separately that we are thereby supplied not merely with two fundamental principles but, into the bargain, with two independent principles. If there is really anything in this duality of principles, we should expect some account of the difference of principle between them. But wherein is the difference supposed to consist? To explain the proposition of identity — *A is what A is* — one slips in underneath it the proposition of contradiction: that *A is what A is* precisely because *A is not other*, and cannot be other, than what *A is*. Now if the principle of contradiction has had to do service in setting

[†] *Translator's note.* The example is puzzling as it stands, since the regular hexahedron is simply the cube. Nielsen's point is presumably that these are questions whose answer cannot be read off the words — unlike "black is not white" — so that intuition must be consulted before one can say whether a contradiction is present at all.

up pure identity — and that, in the logical sense, before it was even born — then it is no more than fair that the principle of identity, once born and the logical birth-pangs happily over, should in its turn help the principle of contradiction to due recognition. Why, then, is no thing what it is not? Answer: because every thing is what it is. And why is every thing what it is? Answer: because no thing is what it is not. In other words: when identity is asked after, one answers with contradiction; and when contradiction is asked after, one answers with identity. On closer inspection it must undeniably strike sound human understanding that there are here not really two fundamental principles but only one.

Yet suppose that, in order so far as possible to avoid conflict with traditional forms, we let formal logic keep its two fundamental principles undiminished — so that, for the sake of “certainty” and “truth” and “determination,” “statuation,” “whatness,” “decision,” “unanimity” and “concordance,”[†] — we held identity literally apart for itself and contradiction apart for itself: would such a holding-apart then perhaps help sound understanding to a clear and distinct insight into the relation between the contradictory and the impossible? At first glance it seems that some difficulty still remains. For if every *A*, provided only that it is what it is, has by the principle of identity a claim to “a valid is,” then it looks almost as though the senseless and the thoughtless could likewise make a claim — a valid claim — to “a valid is”; if every thing, provided only that it is not what it is not, can satisfy the principle of unanimity and concordance, why should the conflicting, the absurd, the self-contradictory and the foolish not be able to satisfy the principle of unanimity and concordance as well?

But to overlook the principle of contradiction, still less to reject it, will not do at all; on the contrary, when thought attempts to deny the validity of the principle of contradiction, it is led by the very denial to confirm it. To posit “black is white” instead of “black is not white” is not only possible but reasonable and necessary if the corresponding contradiction is to be made properly plain; yet in being branded in this way as a contradiction, the contradiction vindicates the principle. The unthinkable, in this sense, is not a something that thought simply cannot think; the unthinkable is the thought which thinking itself condemns, in view of a purpose given within a given context — a thought that comes into being only to vanish, a thought in whose sublation[†]

[†] *Translator’s note.* The list is ironic: Nielsen is parading the scholastic-sounding coinages under which formal logic defends its two principles. *Hvadhed* is a Danish calque of *quidditas*, “whatness”; *Statuering* is the laying down of something as established; *Eenstemmighed* and *Samstemmighed* are “unanimity” and “concordance.”

[†] *Translator’s note.* *Ophævelse*, Hegel’s *Aufhebung*: at once cancelling, preserving and raising up. Where the context is plainly non-technical, the ordinary “cancel” is used instead.

thinking recognizes its own law and satisfies itself. Only when the unthinkable is thus determinately set against the thinkable can the latter be called the logical mark of the possible and the former that of the impossible. Contradiction, then, logically regarded, cannot be abolished outright. Could the principle of contradiction abolish contradiction once and for all, it would at the same stroke render itself superfluous. But since thought cannot stir without contradiction — its sting and its spur — stirring too, the principle stands in force because the cooperation of contradiction stands in force; in order to determine the thinkable, sift the possible and dismiss the impossible, the principle of contradiction must incessantly contradict contradiction, must in resolving contradiction uphold contradiction.

In the application of the principle of contradiction the difference between critical and dogmatic philosophy comes into view. Agreeing with dogmatism in setting contradiction outwardly hostile against thought and the movement of thought, critical philosophy then diverges from the dogmatic in this: that in the unthinkable it sees only the subjective side of the impossible, and not its objective side as well. When Kant attempts to think the origin of the world, he strikes, quite rightly, upon contradiction; the same recurs when he tries to think the essence of substantiality, of necessity and of freedom objectively. The acute thinker, consistent from his own point of view, is here so far from being rid of the contradictory that on the contrary, seeing contradiction tie its knots and settle into antinomies that cannot be waved aside (Propæd. pp. 29–30; Mathem. og Dialekt. pp. 125–126), he declares it insuperable. The objective essence of the world belongs therefore, according to Kant, to what is for us unthinkable; but what does Kant infer from this? By no means that the reality of an objective world should on that ground be held impossible — that is an inference which first arises with Fichte (Propæd. p. 33) — no, the author of "Kritik der reinen Vernunft" warns precisely against precipitate inferences from the subjectively unthinkable to the objectively impossible. The separation of thinking and being, the original characteristic mark of critical philosophy, rests precisely on the presupposition that for the human understanding there are essential, necessary, insoluble contradictions.

But are the Kantian antinomies really insoluble? That depends on whether one chooses to grant contradiction a merely subjective significance, or a significance at once subjective and objective. Such a choice cannot of course be arbitrary; it must follow from the nature of the case. If, then, it turns out that by denying the objective validity of contradiction, holding fast to the critical separation of thinking and being, and so rejecting every attempt to know "*the thing in itself*," one by no means escapes

contradiction but rather entangles oneself in new and greater contradictions — what then?

C.

Dialectic.

What, then, is philosophy? “A doctrine of ambiguity, a science of the ambiguous.” The correctness of this definition seems at least to find ample confirmation in what has gone before. For although we did distinguish between acumen and over-subtlety, we had to admit that one and the same thing could be taken both as over-subtle and as not over-subtle; when it was a matter of separating the thinkable from the unthinkable, what came out was that the unthinkable was something that both could and yet could not be thought; and when it finally came to a resolution of contradiction, the upshot was that contradiction both could and could not be resolved. Should sound understanding find such a predilection for the ambiguous suspicious, let it be noted that the expression “doctrine of ambiguity” is itself ambiguous. While higher instruction in making the clear obscure, the distinct veiled and the intelligible unintelligible is always to be sought among the sophists, philosophy must contrive to set the ambiguous in another light — a light that makes the ambiguity vanish. As the science which everywhere discovers the two-sided in the ambiguous,[†] and by means of contradiction clarifies the doubling, philosophy is *dialectic*^{*)}.

The truth that philosophy by its higher nature is and must be dialectic was categorically stated by Plato. (Cf. *Propædeut.* pp. 124–129; p. 150 and elsewhere.) In Plato the presentation of the dialectical (διαλεκτική from διαλέγεσθαι) has, as is well known, taken the form of conversation; and connected with this is the fact that the dialectician is characterized precisely as the one who in conversation knows how to ask and to answer

[†]Translator's note. *Tvetydig* and *tvesidig*: literally “two-meaninged” and “two-sided.” The definition depends on the Danish pair, which English cannot reproduce.

^{*)}That the form of philosophy is in its essence dialectical must not be misunderstood as though the development of concepts had everywhere to display immediately the formal antinomies by which logical transitions are motivated; on the contrary! Philosophical exposition can and should also take up reporting, descriptive and critical components. The only decisive point is that the elements communicated in the form of report, description, critical discussion and the like should everywhere show themselves, by the design of the whole and the prevailing current of thought, to serve the dialectical. There is in the Hegelian method a formal overstrain, a logical over-exertion, brought on chiefly by the endeavour to have the exposition sparkle at every point with dialectical flashes of lightning. It is with dialectic's flashes of lightning, and with the elements in which they are prepared and out of which they develop, as it is with natural lightning: the lightning-fire must as a rule presuppose clouds, but the clouds are not on that account lightning.

rightly (Τὸν δὲ ἐρωτᾶν καὶ ἀποκρίνεσθαι ἐπιστάμενον ἄλλό τι σὺ καλεῖς ἢ διαλεκτικόν; *Krat.* 390 c.). That what essentially matters in dialectic is not the conversation but the development of concepts conditioned by “the art of language” goes without saying; but that the conversational form, when everything irrelevant is kept out, is nonetheless a very natural form for the beginning dialectician, may readily be shown by a few examples.

Something and Other. A. If the categories “something” and “other” simply excluded one another, there would be no dialectic; if they coincided without more ado, there would be no dialectic either. The dialectical shows itself precisely in this, that the two propositions “something is what it is, not an other” and “something is what it is not, hence an other” not only cancel but also mutually presuppose and ground one another. B. If the dialectical rests on so slippery a ground, then all dialectic is false. A. How so? B. The claim you have set up stands in open contradiction with sound human understanding, and *a philosophy that stands in open contradiction with ‘sound understanding’ is false*. A. If what you here call *open contradiction* is to be dismissed out of hand by an immediate verdict, philosophy can attach no weight to the ruling; for *a philosophy that conforms itself immediately to ‘sound understanding’ is superficial*. But if “sound understanding” will condescend to an inquiry, the inquiry can easily be got under way. B. It would have to be the second proposition, under whose negative form some ambiguity or other has hidden itself, and which might therefore become an object of inquiry; in the first proposition, the proposition of identity, the pure proposition of certainty (*principium certitudinis*), there cannot be a trace of ambiguity. A. But suppose it is after all the first proposition in which the ambiguity, the indeterminacy, and so far the uncertainty, lies hidden — what then? B. Then it must be capable of proof. A. Is it altogether forbidden in logic to say anything else of something than that it is something? B. By no means; for how, by merely repeating “something is something,” should one ever determine *what* something is? Logic, however, which has no fondness for ambiguities, draws a definite distinction, as easy to grasp as it is unmistakable, between determining something by means of an other and saying of something something other than what it is. A. So it is permitted to determine what something is by determining what it is not? B. Not only permitted but necessary. A. Then it is necessary that in the determination of something a determination of an other be contained. B. That too may be granted — but on the express condition that something, precisely by that determination, excludes the other. A. Something, then, is —? B. A thing different from an other. A. And if now, on that condition, we exclude the other, something is —? B. A different thing.[†] A. But now the

[†] *Translator’s note.* *Et Forskjelligt*, the substantivized neuter adjective: “a differing one,” something whose whole character is to be different. English has no natural form for it; “a different thing” is used

other, on its side, is —? B. A thing different from something. A. And when, likewise on that condition, we let the other exclude something, we say that the other is —? B. A different thing. A. So then: something, when determined, is a different thing; the other, when determined, is a different thing; something is therefore what the other is, namely a different thing. B. This ambiguity is easy enough to see through. Certainly something is a different thing, just as the other is; but different and different are two things — something is a different thing precisely by being different from the other, and the other a different thing by being different from something. Something and other are therefore still, as before, different from one another. A. Certainly something and other are different from one another; only a sophist would take it into his head to deny their difference. What was to be proved here was only this: that their difference is grounded in their identity. B. What is different is not identical, and what is identical is not different: only by pure categories can one secure oneself both against hidden ambiguities and against open contradictions.

A. But if I am not mistaken, you have already yourself, by your definite and lucid answers, made valuable contributions toward sharpening our eye for the dialectical unity of something and other. B. I should very much doubt it. A. But you did stress that something can be determined only as a different thing, and, for clarity's sake, as a thing different from an other? B. Yes, that I did stress. A. Then you have at the same stroke expressly required that something, in virtue of that difference, must be taken as an other than that from which it is different. B. Of course. A. So on your view something is other than the other, in opposition to the other — and hence, after all, an other. B. Your play on words leads nowhere; for when something is taken as an other, it becomes another other — namely the other that something itself is, as against all the other that something is not. A. That I can understand; now you are talking like a true dialectician. So you have now seen for yourself that, with respect to the determination contained in something, there are two kinds of other: an other that something itself is, as against another other that something is not — in other words, that something can exclude the other only by itself being an other. B. Yes — so long as one keeps to expressions as general as “something” and “other,” one can no doubt practise without much difficulty the game of thought of turning something into other and other into something. But let us try it with something determinate! Show me in definite examples that something really is identical with other, and I shall grant you that there is “sound understanding” in the dialectical. A. You take a leaf in a book to be something determinate? B. Naturally. A. And the two sides of a leaf

throughout the exchange, and the reader should hear the emphasis on *different* rather than on *thing*.

are surely so opposed to one another that if we call the front a something, we must call the back an other. B. Excellent! Now surely it will occur to no one to identify the front with the back, and then maintain that the front is what the back is, because both are sides, or because the front becomes the back when the leaf is turned, and the back the front. A. But on this occasion too the identity is surely as conspicuous as the difference? B. I can see no identity. A. What do you think, then, about the leaf? B. That the leaf which has the two sides consists precisely in the difference and opposition of the sides! For if the two sides were turned into one side, we should get only a surface, not a leaf. A. And if the two sides were separated in such a way that each came to belong to a body of its own, we should get two surfaces but no leaf. B. Yet it makes an essential difference whether I say "in the leaf the front is united with the back" or "the front is what the back is, the front is back." The first I can grant, the second not. A. If the leaf that unites the two sides were a third thing subsisting on its own over against the sides, we should certainly not be entitled to take it as the existing unity of the sides; but since the leaf, taken in separation from the sides, does not exist at all, the one side must in the leaf show itself identical with the other. B. Are we then perhaps to derive from this that the front itself is both front and back, and the back in turn both back and front? A. That is a consequence you will hardly reject. If we call the two sides α and β , it is evident that α , which in and for itself is neither front nor back, must by its opposition to β be front in so far as β is back, and conversely; α is therefore, in virtue of its identity with β , both front and back. B. That is a peculiar extension of the concept of identity. What I have to object against the concept of identity so extended I would rather hold back, however, until you have produced one more example of a different sort.

A. What do you say to calling the greater a something and the lesser an other? B. That example I find very suitable. A. And what is it, then, that we are to prove? B. What else than that something is both a greater and a lesser, the other both a lesser and a greater, and one and the same magnitude itself both lesser and greater. And in truth nothing is easier than to prove all this. A cursory comparison of three magnitudes, $\alpha > \beta > \gamma$, tells us at once that α is a greater in so far as β is a lesser, and β a greater in so far as γ is a lesser. But what follows from this? That β is in one respect a lesser and in another respect a greater — in that there is no contradiction; but if anyone maintains that the magnitude β is at one and the same time and in one and the same respect both greater and lesser than itself, he does fall into contradiction, and not only with the natural understanding but with true logic. That one and the same side of the same leaf can at one instant be front and at another back, and at one and the same instant — when, mark you, it is regarded

from two opposite points of view — both front and back, contains no contradiction, but a confirmation of the old rule *diversi respectus tollunt omnem contradictionem*. If, on the other hand, one ignores the *diversi respectus*, overlooking what both Plato and Aristotle, for all the difference of their intellectual bent, so strictly inculcated — that in such inquiries what is asked about must each time be seen to be τὸ αὐτό, and regarded κατὰ τὸ αὐτό, πρὸς τὸ αὐτό, ἐν τῷ αὐτῷ χρόνῳ — then one may indeed dazzle the inexperienced for a moment with piquant turns of identity and surprising contradictions, but one will never convince the expert, never satisfy the sensible thinking that inquires after truth. A. Weighty words, and important! So now your “sound understanding” has almost got the better of contradiction and finished off dialectic; I say *almost*, for one small point still remains — a point we simply must have cleared up, and one you will surely be able to clear up easily: it concerns *diversi respectus*, and you have of course studied *diversi respectus*. B. Studied? It has never occurred to me that there was anything further to study in logic’s *diversi respectus*; I have always supposed it enough to remark in a general way that there are *diversi respectus*. A. Then it is surely necessary to distinguish between *respectus*, since there are, after all, *diversi respectus*. Of the two sides of the same leaf — to return to one of our examples — it undeniably holds in one respect that they are the same, namely that both are sides, and in another respect that they differ, namely that one is front and the other back; in one respect the same, that both are, say, written on, in another respect different, that one is, say, white and the other red, and so forth. Each of these different respects may in turn give occasion for new ones; for although it holds of both sides in one respect that both are written on, yet on this very point it also holds in another respect that they differ, namely that the one is written in black ink and the other in red. These doublings of respect can be continued indefinitely. If thinking is not to lose itself in an unsurveyable tangle of all manner of regards to all manner of respects and remarks and considerations and reflections, logic must necessarily distinguish not only between essential and inessential respects but also between respects that fall indifferently outside one another and respects that are contained in one another and therefore follow from one another with inner necessity. That one side of a leaf is written on does not of course entail that the other must be so too; one can take account of the former without taking account of the latter; the former can be without the latter’s therefore being, and no one but sophists would think of positing the latter as identical with the former. It stands otherwise, however, with being front and being back: here it is impossible to take account of the former without taking account of the latter, for these two modes of existence are contained in one and the same existent in such a way that the regard to

the former actually coincides with the regard to the latter — the former is, despite the difference, indeed in virtue of the difference, *identical* with the latter. True insight into the essence of identity, difference and contradiction depends on analysing the *diversi respectus* sufficiently.

B. But can any well-founded objection really be made against the proposition *A* is what *A* is? A. Among others, the objection that the proposition lacks a ground. B. But when the proposition is one with the principle — the principle of identity, the principle of truth, the principle of certainty, *principium certitudinis* — and is of course self-evident in every way, it needs no ground. A. Is it then immediately evident from the proposition *that A* actually is? B. No, not *that A* actually is, but *what A* is. A. Is it immediately evident *what A* actually is? B. Not exactly *what A* actually is, but that, *if A* is at all, *then A* is *what A* is. A. But suppose *A* is not at all? B. Then it is still what it is. A. So in order to be what it is, *A* does not even need to be? B. You understand very well what I mean. A. The question here is not of opining but of knowing; still — what is your opinion? B. That even if *A* is a non-being, it is still what it is, namely a non-being. A. But non-being is surely the same as what we otherwise call nothing, no thing? B. Naturally. A. So then, whether I say "no thing is what it is" or "every thing is what it is," it must at once be evident to sound understanding — by the principle of identity, of course, the principle of truth, the principle of certainty, *principium certitudinis* — that with both propositions I mean one and the same thing!

B. But can anything at all be adduced in defence of so self-contradictory a proposition as "something is what it is not"? A. Among other things this: that a proposition which by its open contradiction so boldly challenges thinking must awaken reflection and call for grounds, in that it demands a refutation. B. But how should one refute by grounds a proposition that immediately refutes itself? A. The proposition "something is what it is not" by no means refutes itself immediately. B. If what is unthinkable, absurd and senseless in this proposition is not self-evident, then what is self-evident, and what is sound human understanding? A. Sound human understanding by no means sees anything absolutely absurd in something's being able to be what it is not. B. So sound human understanding is supposed to be able to conceive that *something* — mark you, at a given time and in a quite definite sense — *is what it* — at the same time and in the same sense — *is not*; no, that really is too much. A. Yes, that really is too much; but so absurd a demand is made by the proposition *something is* — mark you, when its existence is grounded — *what it is not* only when the proposition is held one-sidedly in superficial, abstract immediacy. B. Well, that is another matter, if your defence rests

on an interpretation. A. And another matter too, presumably, if your refutation rests on an interpretation. B. Then prove to us, by means of your dialectical interpretations, that one and the same something can at one and the same time and in one and the same respect be both what it is and what it is not, and that one and the same magnitude can at once be both greater and lesser than itself. A. Such proofs belong to sophistic, not to dialectic. B. But to prove that something can in one respect be what in another respect it is not, or that one and the same magnitude can in one relation be a lesser and in another a greater, nothing is needed but *diversi respectus*, no dialectic at all. A. But to prove that something, though different from the other, is nevertheless essentially one with its other, because the being of both has the same source, the same root; to prove that the same magnitude is on one and the same ground both a greater and a lesser, since the two relations, the two respects, have one and the same point at which they meet, namely being a magnitude; to prove that the identity of something and other, of the greater and the lesser, contains precisely the ground of something's being in existence different from the other, and the greater different from the lesser — for that, dialectic is needed; for this dialectic can prove, but dialectic alone.

Concept and Existence. C. A dialectical development of the inner connection of the *a priori* concepts, and of their transition into one another, is at the present standpoint of the science no doubt to be acknowledged on the whole; but it is another question whether, because the dialectical plays a considerable part in philosophy, one is on that ground entitled not merely to reproach formal logic with not being dialectical, but even — on the strength of a Platonic expression which can always be taken in more than one sense — to transform the whole of philosophy into sheer dialectic. A. Had you followed the preceding inquiry attentively, you would, as to the first point, easily have convinced yourself that there was no thought here of delivering a detailed critique of logical formalism, but only of rebuffing the self-sufficiency with which one not seldom allows oneself, invoking “sound human understanding,” to consign every logical objection to the immediate validity of immediate first principles to the realm of fancy and sophistry; as to the second point, your misgivings could do with a clearer statement. C. *A priori* concepts — such as something and other, essence and semblance, ground and consequent, and the like — can easily be produced out of one another by the application of contradiction. Whoever posits the ground already implies the consequent; the ground is therefore, at bottom, both itself and the consequent.[†] But since the ground becomes a ground proper

[†] *Translator's note.* Danish *Grund* carries the whole weight here: it is at once the logical ground (*Grund* as against *Folge*, consequent), the “bottom” of a thing, and the “reason why.” Nielsen plays on all three, and the English can only follow at a distance.

only on account of the consequent, the ground is consequent and the consequent ground. The same thing is not merely ground in one respect and consequent in another; the two respects also coincide in a single respect: to dialecticize[†] the concepts out of one another is, if I may use the expression, the same as to dialecticize *diversi respectus* into one another. As for the *a posteriori* concepts, the concepts of experience, it is impossible to dialecticize existence out of the concept; or who would try to dialecticize the actual horse out of the concept of a horse, or the actual snail out of the concept of a snail? But if it is impossible to dialecticize the existent out of its concept, then presumably it is for the same reason impossible to dialecticize one concept of existence out of another — the concept of a horse, say, out of the concept of a fly, and this in turn out of the concept of a snail. In the world of experience *diversi respectus* fall outside one another, and where *diversi respectus* fall outside one another dialectic plays a very subordinate part. A. But if you could not succeed in dialecticizing existence out of the concept (cf. Propæd. p. 31), have you never tried going the opposite way — dialecticizing the concept out of its existence, as for instance dialecticizing your concepts of snails out of their corresponding snail-existence? C. One does not dialecticize concepts of experience out of existing objects; one *abstracts* them. A. So there is nothing one might call a dialectic of abstraction? C. Not so far as I know. A. But the concepts abstracted from the objects must surely be, not only in our consciousness, but also in the objects themselves? C. Naturally; for if the concepts were not in the objects, they could not be abstracted from the objects. A. But the object from which the concept is abstracted is surely not itself a concept? C. By no means. A. Yet the concepts, in so far as they are in the objects, may be said to exist? C. That the concepts of experience exist in the existing objects, no one can well doubt. A. So in one and the same existence there are two existents — the existing object, and the concept existing in the object? C. No, it cannot be taken that way; it is not the object itself but its material that is opposed to the concept. A. Then presumably we get three existents: the existing material, the existing concept, and that existence of the object in which both exist? C. No, one cannot take it that way either. A. How, then, is it to be taken? C. The material by itself is no separate existent, any more than the concept is; the material and the concept are two sides — two sides of existence — of the same existing one, the object. A. The existing one is, then, over against and in opposition to the two sides of existence, the existent in and for itself? C. No, one cannot take it that way either; for if we let abstraction remove the two sides of existence, the

[†] *Translator's note.* *Dialektisere*, Nielsen's own verb and deliberately ungainly in Danish too: to derive one thing out of another by dialectic. It is kept rather than paraphrased, since the whole dialogue turns on the question of what can and cannot be so derived.

existing unity will itself become an abstraction, a side of existence that does not exist. A. But how, then, is it to be taken? C. So that the existing unity is seen to exist in the two sides of existence, which precisely thereby come to exist in inseparable unity. A. Then however many parts, qualities and sides the existing object may otherwise have, we may now, on your account, assume that these parts, with their sides, qualities and mutual relations, are all equally material by virtue of the material and intellectual by virtue of the concept; every side of existence thus becomes an existential unity of two sides, the material and the intellectual. C. That we may quite safely assume. A. But what is material in the object is surely something sensuous,[†] which as sensuous cannot be thought, and what is intellectual in the object something non-sensuous, something thinkable that cannot be sensed? C. Perfectly correct. A. Then in the existing unity the unthinkable is after all a thinkable, and the non-sensuous a sensuous. C. That I do not understand! A. Then neither do you understand the existing unity; for how should the thinkable in an existing unity be able to penetrate the sensuous without the sensuous thereby becoming, by virtue of the thinkable, itself thinkable, and the thinkable, by virtue of the sensuous, sensuous? C. It almost looks as though there were something in what you say. A. The concept of existence, then, is a system not only of intellectual but also of material determinations — a system of intellectual sense-determinations, of sensuous intellectual determinations. C. Yes, so it is. A. But then the concept of existence betrays a most striking contradiction. C. What contradiction? A. That of demanding existence by setting itself against existence. C. What does that mean? A. If the whole system of sensuous intellectual determinations is posited in the intellectual form, then the sensuous vanishes as a moment of reflection sublated in the thinkable — that is to say, the existent transforms itself into the concept of existence, but existence goes out. If conversely the whole system of intellectual sense-determinations is posited in the sensuous form, answering to immediate sense-impressions, then the object steps into existence, but the concept of existence goes out. . . . C. No, the concept of existence does not go out; it is completed in existence: the existent is, by what we have said, nothing other than the concept of existence complete in every respect. A. But was it not you who, by what we have said, granted that the object from which the concept is abstracted was not itself the concept? C. Yes, but I was thinking then of the abstract and so far incomplete concept, not of the concept concrete and complete in itself. A. So it is after all possible to dialecticize existence out of the concept, once one has learned to dialecticize the concept out of existence? C. To prove that in deed would here amount to showing it

[†] *Translator's note.* *Sandselig* = *sinnlich*, pertaining to the senses; it carries none of the modern connotation of "sensual."

in definite examples.

A. Sound human understanding tells us that the horse is an animal, that the fly is an animal, and the snail likewise; but how do such judgments actually stand to “the principle of identity and of contradiction”? C. The matter is plain enough. A. Yes, the matter is that if we let formal identity hold for the predicate, we deny it in each subject; whereas if we let it hold for each subject, we deny it in the predicate. C. How so? A. If we posit unconditionally “animal is animal,” then the horse becomes a fly and the fly a snail; if we posit unconditionally “the horse is horse, the fly is fly, the snail is snail,” then animal (horse) is not animal (fly) — that is, the common predicate contradicts itself and must fall away. C. This difficulty can be overcome well enough. A. With dialectic or without? C. Without any dialectic at all! One need only bethink oneself of the factual characteristic marks of the concepts of existence, and of the relation of the disjunctive judgment to *diversi respectus*. A. That we must certainly hear! C. If we compare animals with plants, it is seen that all existing animal individuals have one thing in common: being animal. If on the other hand we compare the animal existences with one another, then one animal series shows itself, by given characteristic marks, different from another. If we compare the existences in the one series with the existences in the other, then the individuals within each series are alike; but if instead we compare the existences belonging to a given series with one another, particular marks will show how one class is distinguished from another, and so forth. The animal, then, in opposition to the plant, is what the animal is; but by general and constant marks of difference the animal, in opposition to itself, is either mollusc or articulate or vertebrate.[†] In opposition to articulates and vertebrates the mollusc is always mollusc; but in opposition to itself the mollusc is either a mollusc proper or a radiate. Now if one simply goes on in this way with the divisions (*divisiones* and *subdivisiones*) through every series, every class, every order, every family, guided by the marks established by experience, until by way of genera and species one has reached individuals — what contradiction is there in that, and what have such classifications to do with dialectic? A. Unless it be that dialectic finds a good deal to do with *diversi respectus*. C. There is surely no danger of that. A. Then tell us, on the strength of your completed classification: what actually is a horse? C. A horse (*Equus Caballus*) is, by the category of its kingdom, an animal; by the category of its series, a vertebrate; by the category of its class, a mammal; by the category of its order, a pachyderm; by its

[†] Translator’s note. *Bløddyr*, *Leddyr*, *Hvirveldyr*, and *Straaledyr* just below, are the Danish names of Cuvier’s four *embranchements*: Mollusca, Articulata, Vertebrata and Radiata. The classification was standard in 1861 and has since been abandoned; “radiate” covered what we should now distribute among the echinoderms, cnidarians and others.

family category, akin to asses and mules; and a species-category of its own within a genus-category of its own. A. A fine collection of categories, then, all of which had to combine here to yield the horse's concrete category of existence! If only one could now, with the help of *diversi respectus*, get it properly explained how these many different categories have actually grown together — not to say, been bundled together — into a single one! C. They of course enclose one another in such a way that the more general always encompasses the more special. A. Surely not, though, in the way one envelope encloses another, or one ring encircles another? C. No, they rest upon one another, the more special having in each case settled upon

the more general. A. The more general a category is, the more abstract it is; the more abstract it is, the further it is from existing: so the existent rests upon the non-existent? C. No, I meant to say: they rest upon one another in such a way that the more general is always borne by a more special, this by one more special still, until we come to the most special of all, which is one with the existent itself. A. So it is not being a horse that depends on being an animal, but being an animal that depends on being a horse? C. No, that is not how I conceived it either; indeed, at this moment I cannot, to speak plainly, say how I actually did conceive it.

II.

Philosophy and the Particular Sciences.

General Remarks.

“With respect to the a posteriori, the advance of knowledge will doubtless depend on new information’s being constantly gathered, new matters of fact tested, new discoveries made, and so on; but is the matter thereby settled? If the advance of knowledge in relation to the a posteriori is regarded from this side alone, it is easy to dismiss philosophy. If it cannot so much as exhaust the a priori — seeing that mathematics owes it no thanks — how much less can it meddle with positive matters of fact! Let the historian lay a problem before it: it cannot answer, for a historical problem is answered only historically. Let a chemist apply to it: it has no knack for experiment and makes no discoveries in chemistry. Let the astronomer want help with an observation: it scarcely understands him. Let the physician ask it for advice: it is ignorant. Even what it has speculated out on its own account concerning the relation of soul and body the physiologist may unfortunately be unable to use.” (Propædeut. pp. 99–103.)

The extent to which it holds that philosophy is dialectic is seen from its relation to the subject-sciences,[†] for that relation is dialectical in the strictest sense. If one overlooks the dialectical, the relation becomes ambiguous and doubtful; for while the subject-sciences are by nature inclined to dismiss philosophy, philosophy in its turn, when it conceives its vocation dogmatically, is tempted from its side to exalt itself at the expense of the other sciences. Schelling’s words are characteristic in this regard: “Speculation is everything;

[†] *Translator’s note.* *Fagvidenskab*, literally the science of a *Fag* — a subject, a trade, a department. “Subject-science” is used here, as in the translations of Nielsen’s other works, and its practitioner, the *Fagmand*, is “the specialist.” These are to be kept distinct from *de særskilte Videnskaber*, “the particular sciences,” of this part’s title.

it is to behold in God, to contemplate that which is. Science itself has worth only in so far as it is speculative, that is to say, contemplation of God as he is." (Propædeut. § 17, pp. 81–83).

What has chiefly contributed in recent times to confusing the issue is the circumstance that Hegel — whose penetrating grasp of contradiction, of the negative, and of the immanent method of development (Propæd. pp. 150–158) must place him so high in the ranks of the dialecticians — has nevertheless, in his account of philosophy's relation to the subject-sciences, overlooked the dialectic of the relation. Instead of acknowledging the scientific independence of the particular sciences, Hegel closes his propaedeutic survey of the encyclopaedia of the sciences (Propæd. pp. 84–85) with the declaration that the encyclopaedia is properly "an exposition of the general content of philosophy; for whatever in the sciences is founded upon reason depends upon philosophy, whereas whatever there is in them of what rests on arbitrary and external determinations, or, as it is called, is positive and statutory, lies outside it." But to let philosophy's superiority rest in this way directly upon the subjection of the subject-sciences is to take the matter undialectically; to see philosophy's independence grounded in its dependence upon the independent subject-sciences is, by contrast, to grasp the relation dialectically.

The independence of the subject-sciences over against philosophy is evident from the course of their development; for the methods and discoveries that mark the great advances of these sciences are so far from being attributable to philosophy that "the history of the inductive sciences" (Whewell) shows, on the contrary, to what degree independent research and a procedure answering to the nature of the case have depended on the researcher's being able to free himself from the one-sided influence of philosophical speculation (Propæd. pp. 167–68). What in the subject-sciences "is founded upon reason" by no means depends upon philosophy in particular. In the first place it is not reason that owes its origin to philosophy, but on the contrary philosophy that owes its origin to reason. And next — and in this connection this is decisive — it must be emphasized in the strongest terms, not only for the sake of the subject-sciences but above all for philosophy's own sake (Propæd. § 23), that there is a knowledge which is scientific in the strictest sense although it is not philosophical; indeed a thinking whose scientific energy depends precisely on its not being philosophical. For that in the empirical knowledge of the empirical there is a certainty of sensation, a many-sidedness of representation, an added gift of intuition which no philosophical glimpse of the Idea can replace, is surely as undeniable as it is undeniable that there is an exact and critical thinking not to be exchanged for the dialectical (Mathem. og Dialekt. p. 3). The rational distinctiveness of

the particular sciences is seen, too, from the fact that each one severally presupposes a scientific talent of its own answering to it.

But if one now sets up the proposition that *the subject-sciences are independent of philosophy, whereas philosophy — particularly as regards the appropriation of positive content — is dependent on the subject-sciences*, then this presupposition, true as it is in itself, may easily occasion a false inference. If philosophy, it is said, is so dependent on the exact and positive sciences in everything that concerns the exact, the empirical and the positive, then this dependence must plainly be a sign of its lack of independence. Every time philosophy comes into contact with some determinate positive subject matter, be its content mathematical, mechanical, chemical or historical, the philosopher must of course put up with being not merely set right about the facts but even duly schoolmastered by the specialist in the development of the concepts themselves. It looks, then, as though the concepts were perfectly trustworthy, clear, distinct and secure in all their consequences so long as they are preserved in the form in which the subject uses them, but abandoned to distortion as soon as they are made an object of philosophical reflection; it looks as though the philosopher had to borrow everything from the specialist, while the specialist for his part may regard the philosophical as a matter of complete indifference. This way of seeing things rests on a misunderstanding. Philosophy's dependence on the subject-sciences is, essentially regarded, a sign of its independence, of its ideal self-validity.

In contrast to the material objects of the surrounding world all sciences are of course ideal, all have ideality; but in contrast to philosophy, as the representative of ideality, the subject-sciences must be said to stand for reality. If the opposition is determined categorically, the real in its finitude will be recognized by its asserting itself through excluding its other, whereas the ideal proves its reality precisely by taking the other up into itself and relating itself to itself through the other. That mathematics as mathematics, chemistry as chemistry, natural history as natural history each keep to their own^{*)}, and, even when contributions are fetched from other subjects, still have first and directly to do each with its own, is precisely a mark of these sciences' finite reality. That philosophy, by contrast, must always have its ideal wealth in real poverty, its positive in its negative, was illustrated by Plato in an ingenious myth: at Agathon's banquet Plato has his Socrates, as is well known, present Eros, the ideal philosopher, as the son of Poros and Penia.

^{*)} "There is a cardinal rule which can never be violated with impunity, and which is nevertheless constantly violated in our groping advance toward the light. It is: *never to attempt the solution of the problems of one science by the order of ideas belonging to another*. There is one class of conceptions belonging to physics, another belonging to chemistry, a third belonging to physiology, a fourth belonging to psychology, a fifth belonging to social science. While all these sciences are closely connected, each has its own independent province, which must be respected." G. H. Lewes: *Hverdagslivets Physiologie*, Danish translation, part 1, p. 66.[†]

The realistic science, once founded, may in the course of time need manifold reshaping and undergo many changes, yet its supports and foundations will through it all remain unshaken. If the idealistic science, on the other hand, is to remodel its building, a *de omnibus dubitandum* will gradually shake every support; the whole building must then be pulled down to the ground, that it may be raised in its new style from the ground up. Each time such a critical juncture arrives it does indeed look for the moment as though reality were at stake, as though it were now all over with philosophy; but history confirms what also lies in the nature of the case, that such pervasive crises mark precisely the turning points in philosophy's rebirth.

"The subject-sciences are solid; they always have steady work, ample means and secure recognition in the republic of letters. Philosophy, by comparison, is like a traveller without settled provision, a hovering shape that sinks and rises, dies away and lives again, as the spirit will." (Propæd. p. 102). — Yet it is only from a wholly external, realistic standpoint that the ideal is taken as merely the vague, the indeterminate, the in itself indeterminable; the dialectical form is a form determinate and determinable in itself, and philosophy's relation to the subject-sciences is therefore to be stated categorically.

If philosophy is to be able to place itself in an inner and essential relation to a particular science, it must discover in that science a problem which it can also call its own problem. In order to show how such a *common problem* is chosen and treated, and so as not to scatter our attention over too many points, we shall in what follows confine ourselves to examples drawn from mathematics, chemistry and physiology; and we have thus, in mathematics, *the problem of infinity*; in chemistry, *the problem of change*; in physiology, *the problem of life*.

But even if such common problems are characteristic enough when viewed from the side of ideality, it is by no means thereby settled that they show themselves so when they are grasped realistically. Because the philosopher has noted some pervasive problem in one or another determinate subject-science, it is not yet said that the specialist will acknowledge the same problem in a corresponding way. A chemist who wished absolutely to resist a philosophical conception might, for example, reason thus: "There is in chemistry properly nothing at all that one may call a problem of change, but rather a concept of change, which arises and comes to be applied in so far as chemistry investigates the natural laws according to which the inner changes of bodies take place. But it is one thing to investigate the laws according to which the changes take place, and another to want to fathom the essence of change, to speculate about its possibility and impossibility — a task wholly foreign to chemistry." In the same spirit a physiologist

wishing to dismiss philosophy might reason thus: "Physiology does indeed proceed from the presupposition that there is a something we must call life, since in living organisms there is a law-governed system of vital phenomena; but since physiology essentially confines itself to investigating the phenomena, without penetrating into what lies hidden behind them, it can by no means be granted that there is here any proper problem of life. What life is in itself — a principle, a force, a product of the peculiar cooperation of mechanical and chemical forces — no matter! Let each form whatever individual representation of the essence of life he pleases; so long as he is not thereby kept from seeing the validity of the laws and demonstrating the physical causes, physiology is satisfied."

The difficulties which the natural influence of differing standpoints on the conception of a scientific problem always brings with it are considerably heightened by the ease with which something human — that is to say, something one-sided, accidental, arbitrary, passionate — mixes itself into the investigations. And since philosophy, for all its ideality, is like every other science subject to human conditions, it becomes important in this connection to take the matter from the purely human side as well, and so to note the obligations and rights which the sciences have to uphold in their mutual conflicts.

That the common problems in question must look otherwise in philosophy than in the subject-science is a matter of course. The problem of change, even if the chemist will acknowledge it in his own way, is in chemistry by no means what it is in philosophy; likewise the problem of life, even when the physiologist grants its reality, is one thing in physiology and another in psychology. The philosopher who knows the problem in the form of otherness, and is aware that the same is nevertheless not the same, must therefore make it his chief business to clarify the conditions of the development and to make the transformation plain. Anyone who would seek guidance in a particular science must of course make himself at home in its way of thinking. If the philosopher neglects this — if he misunderstands the character of the science in question either as a whole or in essential particulars, whether by ascribing to it an aim it does not have and cannot have, or by tearing its doctrines out of their proper connection and so distorting them, or by overlooking the peculiar presuppositions and conditions under which a given concept with its corresponding designation was formed, thereby laying a false meaning under the concept and in this way coming to force upon the science views, results and consequences it cannot acknowledge — then the specialist is within his rights in speaking out. This right becomes a formal duty to the truth when such blunders on philosophy's part, as has not seldom been the case (*Propæd.* pp. 170–171), are combined

into the bargain with the conceit that it is philosophy which from its exalted standpoint must feel called upon to reform the subject-sciences. But further than a rejection of philosophical encroachments the specialist cannot and should not go. If he goes further — so that, for example, he is not content to secure to the concept the meaning it has in his science, but regards it as belonging to his subject to such a degree that he either flatly forbids the philosopher to touch it, or at any rate takes it upon himself to prescribe how far, in what manner and in what connections philosophy may develop the borrowed concept graciously made over to it by the subject-science — then there is to be seen in this not merely a presumption but a ludicrous misunderstanding.

"Confined to the several compartments of consciousness, the concepts gained are still so grown together with finite constituents, still so weighed down with particular presuppositions, that they find themselves as it were in a state of unfreedom. They are held here as the prisoners of their matter within a determinate horizon; they are, so to speak, the subject's bondsmen, who dare not leave the appointed place of work." (Propædeutik p. 113.)

Now in setting out to free the real concepts from what is accidental in their delimitation, to unfold the consequences and to bring the schematic forms into dialectical flow, philosophy must of course — once it has drawn the concepts into its own horizon, posed the problems in its own way, and thus taken the whole responsibility of the treatment upon itself — have an indisputable right within its own province to apply its own measure, to introduce its own designations and usage, and to carry out the development in its own spirit. If it stands firm that every subject-science must in essentials contain its own correctives, then philosophy too must surely contain its correctives. A judgment upon the errors and blunders a philosopher may possibly have committed in carrying through the dialectical analyses belongs in the last instance not before the forum of the subject-science but before that of philosophy.

To deny philosophy the full scientific right to treat in a free philosophical manner what it has appropriated from the subject-sciences is to deny it a development of concepts of its own, and to resort to a dictum of authority. If it is to go by dicta of authority, then a specialist may settle philosophy's fate as easily as that caliph settled the fate of the Alexandrian library: "Either it contains only what stands in the Koran, and then it is superfluous, or it contains something else, and then it is false"; — "either philosophy contains only what stands in the subject-science, and then it is superfluous, or it contains something else, and then it is false." It is a Saracen dilemma; but a Saracen dilemma cannot make philosophy give up its right.

A.

Philosophy's Relation to Mathematics:**The Problem of Infinity.**

"A philosophical fox or marten has stolen in upon mathematics' fat poultry to fill his schematic belly; he means to suck out their warm blood, so that nothing is left but the dead carcasses, void of sap and strength, which are then to be presented as mathematics' vigorous, practical, lively concepts" (Professor Steen's "Afregning," p. 7).[†]

These boding — indeed ill-boding — words may well seem at first glance to have something particularly forbidding about them, especially when set beside a declaration such as this: *"In the first place there is nothing that one may name 'the problem of infinity' in mathematics"* (cf. Begrebet: Nøiagtighed, p. 38). But when it is thereupon conceded that there is *"a mathematical concept 'infinite' which arises and comes to be applied, and which is extremely important and moreover difficult, because it is of a mixed nature"*, the forbidding element disappears and something inviting shows itself. For if the mathematical concept "infinite" which arises and comes to be applied is, on the mathematician's own admission, *"extremely important and moreover difficult, because it is of a mixed nature"*, then this contains, so far as philosophy is concerned, a well-founded invitation to investigate whether this concept of infinity, in mathematics so *extremely important and moreover difficult*, may not after all be connected in one way or another — perhaps without the mathematician's knowing it — with a problem running through the whole of mathematics; whether philosophy may not have every reason to acknowledge, in this pervasive problem, its own age-old problem of infinity; and whether mathematics, without a mathematician's always quite thinking of it, may not be said to furnish essential and indispensable contributions toward that problem's solution.

a) The Concept of Infinity.

That the concept "infinite" which in mathematics "arises and comes to be applied" is "extremely important" we may grant at once; but we must most definitely deny that the

[†] *Translator's note.* This whole section is a reply in an ongoing quarrel. In 1860 the mathematician Adolph Steen attacked Nielsen's *Philosophie og Matematik* in a pamphlet called *Afregning* ("Settling of Accounts"), and Nielsen answered with two pieces of his own — *Hr. Professor Steens "Afregning": Hermed en specificeret Opgjørelse til Revision* and *Om Begrebet Nøiagtighed med særligt Hensyn paa Hr. Prof. Steens "Bidrag"*, the latter a postscript to *Philosophie og Matematik*. The short titles Nielsen cites here and below — *Afregning*, *Begrebet: Nøiagtighed*, *Om Begr. Nøiagt.* — are to those pieces, and the words he puts in quotation marks are his opponent's objections as he reports them there. Without this the pages read as an argument with nobody.

concept of infinity, taken as a concept by itself, is either difficult or of a mixed nature. Of the infinite in general only this can be said: that it is a negation of the finite, the not-finite. If any difficulty lay hidden in such a negation, the difficulty could not stem immediately from the negation itself, but would have to be referred essentially to what the negation cancels. So if one is to find difficulties in the mathematical “infinite,” one must find corresponding difficulties in the mathematical “finite”; and if one raises no objection to the latter, it betrays mere thoughtlessness to raise objections to the former.[†] If the concept of infinity is difficult “because it is of a mixed nature,” would not the concept of finitude, when the finite is specially emphasized, be of a still more mixed nature? If it be objected that the mathematical “finite,” which holds only of magnitudes and not of things, cannot possibly be of a mixed nature, then one will also be obliged to grant that the negations in which such a “finite” disappears can as little be so.

Concepts of a mixed nature always have a questionable admixture of the conceptless, and as mixture-concepts they stand on the lowest rung in the order of concepts. How does a mixture-concept come about? By parts of unlike kind, brought together in an external and accidental way, being subsumed under a collective unity, which in consequence comes to contain marks of unlike kind accidentally brought together. Thus the concepts “topsoil,” “gravel,” “marl” are genuine mixture-concepts; but a concept of gravel and a mathematical concept of infinity — what a difference!

But, someone may perhaps ask, if the concept “infinite,” in however many and various connections it occurs and however many transformations it undergoes, always preserves one and the same unalterable fundamental meaning — namely that of being “the negation of finitude” — how does it come about that so pure, so negative a logical concept can in a mathematician’s eyes take on the appearance of being “of a mixed nature,” or, what is the same, of belonging among the mixture-concepts? The answer is not difficult: *one mixes the accidental with the essential; by this mixing one turns up a number of meanings of differing kinds; one does not derive these separate meanings from the fundamental meaning, does not exhibit the necessary transition of the one into the other, but sets them up externally side by side, makes them an object of enumeration — and behold, the thing goes as of itself.* If one puts such meanings as “negation in positive form,” “impossible,” “transition involving something qualitative,” and so on, externally together, one gets at once an impression of the mixed. How is one to distinguish a negation in positive form from a negation in negative form, except by the former’s expressing the negative mediately

[†] *Translator’s note.* Danish *gjøre Ophævelser over* is an idiom, “to make a fuss about”; but *Ophævelse* is also the technical word for sublation, and Nielsen has just used its verb (*ophævet*, “cancelled”) two sentences earlier. The pun does not survive.

and the latter immediately? But if now the finite is set against the infinite, it is not the infinite that expresses the negation in positive form but on the contrary the finite, in so far as the finite, though it has a positive form, is yet mediately determined as the not-infinite. To take negation's own immediately negative expression for its positive form betrays a comical confusion. If someone, to display his acumen in matters logical, were to maintain that the proposition " a is not something" was straightforwardly negative, whereas the proposition " a is nothing" was affirmative, the negation being lodged in the word "nothing," one would surely agree without difficulty to assign such trials of acumen — formulated in all clumsiness after a formal logic's S is *Non* — P — if not outright to the pedantic, then at any rate to that class of immature attempts which betrays that the acute man has not yet learned to distinguish between fruitful and unfruitful distinctions. If one now grants, so as not to be saddled with too much logical mixed goods, that the proposition " a is nothing" is just as directly negative as " a is not something," then one will also grant that the proposition " a is zero" is just as directly negative as " a is nothing." But if it stands firm that the sign 0 is just as direct and immediate an expression of negation as the word "nothing," then it takes a quite abnormal acumen to see that, while the proposition "between a and a there is no difference" is straightforwardly negative, the equation $a - a = 0$ should on the contrary be so nailed fast in the positive that only the screw of a paraphrase could bore through to the negation. (Cf. Om Begrebet: Nøiagtighed, p. 39).

The logical construing of mathematical equations is a peculiar business; the more weight is laid here on the conceptual side, the more evident it becomes that philosophy has a word to say here too. On the natural mathematical way of thinking, the equation $\frac{a}{0} = \infty$ shows us the well-known relation between divisor and quotient in its extreme consequence: if, on the assumption of a constant dividend, the divisor goes on shrinking to infinity, the quotient must go on growing to infinity. There is nothing in this relation to recall the impossible; everything points on the contrary to the possible: to the smallest possible divisor there answers the greatest possible quotient. If, on the other hand, in order that the infinite may come to signify the impossible, one chooses another point of view and construes the equation thus — "When it is asked how many times 0 must be made greater in order to give a , the answer in words is: that cannot be produced by multiplying 0; but in signs: $\frac{a}{0} = \infty$, 0 must be made infinitely many times greater in order to give a " (Om Begr. Nøiagt. p. 40) — then such a construal is well fitted to promote all manner of admixtures of an illogical miscellany. If the equation $\frac{a}{0} = \infty$ were really and seriously designed to answer the in itself absurd question "how many times must

0, the absolute 0, pure nothing, be made greater in order to give a ?”, then notations such as $\frac{a}{0} = 0$, $\frac{a}{0} = a$ would have been far more adequate. For from such expressions the impossible, the absurdity in the demand, is seen instantly, whereas the equation $\frac{a}{0} = \infty$, by its ironic pointing to the infinite, only mimically hints at what is absurd in the question, and then — quite rightly — transfixes the questioner by “a negation in positive form,” because he is asking not as a mathematical thinker but as a logical fool. If one gives up the logical foolery of trying to reckon how many times pure zero must be made greater in order to produce a ; if, in order that the equation may yield any sense, one lets 0 denote the smallest possible divisor while ∞ denotes the corresponding greatest possible quotient — then it is seen from every side how the impossible gives way to the possible. From the equation $\frac{a}{0} = \infty$ there follows directly, in virtue of the concept of the relation in question, $a = \infty \cdot \frac{a}{\infty} = \infty \cdot 0$; likewise from $\frac{b}{0} = \infty$, $\frac{c}{0} = \infty$, $\frac{d}{0} = \infty \dots$ there follow $b = \infty \cdot \frac{b}{\infty} = \infty \cdot 0$, $c = \infty \cdot \frac{c}{\infty} = \infty \cdot 0$, $d = \infty \cdot \frac{d}{\infty} = \infty \cdot 0 \dots$ Instead of the impossibility of $\infty \cdot 0$'s being able to give a , we see on the contrary the possibility of $\infty \cdot 0$'s giving not only a but b , c , $d \dots$ — in short, as many different magnitudes as you please. Instead of the curt, dismissive “impossible,” we get possibility upon possibility, possibilities in abundance. This result is of course so far from surprising the *thinking* mathematician that it does not even surprise ordinary sound human understanding. Everyone understands that any finite magnitude whatever may — and this is a possibility — be said to consist of infinitely many infinitely small parts, hence $a = \infty \cdot \frac{a}{\infty} = \infty \cdot 0$, and that it may therefore also — and this is a possibility — be divided into infinitely many infinitely small parts, hence $\frac{a}{\infty} = 0$.

The fixed idea that the concept “infinite,” as a “negation in positive form,” should be better fitted to denote “the impossible” than the concept “finite” is easy to clear up. For “the impossible” is in and for itself as indifferent to “infinite” as to “finite”; the impossible shows itself every time a demand that has been set up annihilates itself, and such self-annihilating demands may just as well presuppose the finite as the infinite. It is, to stay with the infinite, impossible to *enumerate* all the terms of an infinite series; it is, to pass to the finite,

impossible for a human being to *count up to* a quadrillion; and it is — here we must lean at once on both the finite and the infinite — impossible that a sphere with an infinite radius should be a sphere, or a plane a spherical surface with a finite radius; and just as impossible that the equation $a = 2a = \infty \cdot a = \frac{0}{a}$ should have any reality. If in the language of mathematical signs one wants a more direct notation for something impossible in itself, then it is not 0 and ∞ but $\sqrt{-1}$, $\sqrt[n]{-a}$ and the like that one has to keep to. (The

“Afregning” as revised by R. N. Appendix, pp. 26–27).

In logical generality it is easily seen that the concept “infinite” does not as a concept take up constituents of unlike kind, and so cannot be reckoned among the mixture-concepts. That the concept nevertheless occurs in several meanings, shows itself in a different light, and to that extent may, when treated superficially, acquire a tinge of being “of a mixed nature,” has its ground in something quite different: to think the concept through is to unfold the problem.

b) The Problem of Infinity. Suppose someone were to open an objection to a philosophical reference to the way in which the problem of finitude finds its solution in mathematics with the following declaration: “In the first place there is nothing that one may name the problem of finitude in mathematics, but rather a mathematical concept ‘finite’ which arises and comes to be applied, and which is extremely important and moreover difficult, because it is of a mixed nature.” One would surely find that a strange thing to say. What a finite magnitude is, is soon said; but the problems concerning the mutual relations of finite magnitudes are countless. Instead of saying that there is no problem of finitude in mathematics, one would rather have to say that there are countless problems of finitude in mathematics. If these problems fell apart from one another in loose multiplicity, one might indeed deny that there was any general problem running through them all; but one would then at the same stroke have to deny that there was any mathematical method — in other words, that mathematics was a science. If mathematics is a science, if there is a method by which the various problems of finitude are framed and solved in systematic connection, then there is also a general problem running through them all, a fundamental problem embracing and determining the method in all its ramifications, a problem whose all-round solution demands nothing less than the whole of “the analysis of the finite.”

But what holds so incontestably of “the problem of finitude” in its relation to “the analysis of the finite” — should this not hold also of the relation between “the problem of infinity” and “the analysis of the infinite”? What holds so incontestably of the concept “finite” in its relation to “the problem of finitude” — namely that the difficulty lies not in the concept but in the problem — should this not, on closer inspection, be found to hold also of the concept “infinite” in its relation to “the problem of infinity”? —

α) Concept and Problem. As the distinguishing mark of mathematical concepts — however much they may otherwise resemble the general logical ones — it must be expressly emphasized (Mathem. og Dial. pp. 21–27) that they are operational concepts. From this it follows that, while in their generality they present no difficulty, analysing

thought will soon discover that their proper development must always take place in the closest connection with the operations whose logical essence they are meant to clarify. What a logarithm is can easily be stated in a definition exact enough for sound understanding to hold the concept fast; but the manifold and characteristic consequences in which the concept's content first really shows itself come forth only in the course of the operations. What is meant by dividing a line in extreme and mean ratio — or, as it is also called, by golden-sectioning the line — is soon said and easily grasped; the concept "golden-section"[†] presents no difficulty. The difficulty, if there is one, is to solve the problem of the golden section and to develop all the consequences contained in it. What holds of the particular operational concepts holds of the general ones too: only from the systematic solution of the problems of finitude does it come to light what mathematically lies in the concept "finite"; only a carrying-through of "the analysis of the infinite" can open us a scientific insight into the mathematical concept of infinity. From the consequences of the operations it is further seen that the infinite falls not only outside but also inside the finite (cf. *Phil. og Mathem.* pp. 20–24; *Mathem. og Dialekt.* pp. 28–38), and that it develops in manifold ways out of finitude itself. Had a pervasive problem of infinity not originally stirred beneath "the analysis of the finite" and declared itself in many separate flashes, the differential and integral calculus would never have been developed. When Leibniz solved "the problem of tangents," what was essentially at stake was not the tangent and the secant (cf. *Phil. og Mathem.* p. 64) but the articulating reciprocity of finite and infinite: in "the problem of tangents" "the problem of infinity" is to be seen.

β) *The Mathematical Conception.* That thinking mathematicians, however much weight they may otherwise lay on the satisfactory results of the operational method, have nevertheless never been able to separate satisfactorily the concept they find inconceivable from the method that rests upon that concept, is shown by the history of mathematics from Leibniz and Newton (cf. *Mathem. og Dial.* pp. 90–91, p. 95 and elsewhere) down to the present time. "In the first place, when the principle is unknown we cannot know whether all the consequences following from it can be admitted: we go as if blindfold, and it might well happen that after laborious effort we should at last strike upon something which exhibited precisely the nature of the principle — that is to say, upon something we could not conceive. In the second place: is it not a shameful degradation of reason to make it follow a concatenation of terms, a series whose beginning and end

[†] *Translator's note.* The Danish verb is *høidele*, literally "to high-divide," the nineteenth-century Danish term for division in extreme and mean ratio, i.e. the golden section. English has no corresponding verb, so "golden-section" does duty as one here and in the next sentence.

it cannot see through? It is a great evil that is born of that kind of routine: accustomed to operating without seeing the ground, the mind loses all practice in thinking, and ends by carrying everything out quite mechanically.” (Mathem. og Dial. p. III). These misgivings of Savrien’s sufficiently explain the unrest and tension in which the analysis of the infinite has constantly held mathematical reflection^{*)} (Mathem. og Dial. pp. 55–56), whose problem — as a prize question set by the Berlin Academy in the year 1784 characteristically puts it, “wie aus einer widersprechenden Voraussetzung so viele wahren Sätze haben können hergeleitet werden” [how so many true propositions could have been derived from a contradictory presupposition] — will not cease to return, even though “routine,” the mechanical facility, will at all times no doubt afford “*habiles Calculateurs*” satisfaction so ample that it may easily escape this or that reckoning-master’s notice that there is in mathematics something one may name “the problem of infinity.”

γ) *The Philosophical Conception.* “To make the problem of infinity recognizable, reflection must lean upon the operation; to solve the problem, it must appropriate the intelligence expressed in the operation. That there can be a finite ratio between infinite magnitudes; that a sum of infinitely many terms may in some cases make up a finite and in others an infinite magnitude; that a series whose terms are so related that the limit of the ratio of two successive terms gives the quotient 1 may in some cases be convergent and in others divergent, and so on — such things will everywhere escape the thinking that does not, at the operation’s hint, catch sight of the exact determinations. That these determinations must in philosophical form become dialectical, while in mathematical form they neither are nor can be so, shows a characteristic difference between mathematics and philosophy, but by no means any conflict. No concept, however exactly it be defined, can maintain itself in a being indifferent to the other concepts: the validity of concepts is seen in their transitions; but the transitions which exact analysis makes by the operation, the thinking that penetrates the operational concepts makes by dialectic” (Mathem. og Dial. p. 57. Cf. pp. 59–60). If the finite is set immediately against the infinite, sound understanding involuntarily gets the impression that it is the finite which, in its multiplicity of terms and relational determinations, contains the wealth of existence, whereas the infinite is a bottomless nothing, waste and empty as empty space. And since thought can determine only what is in itself determinable, and only that is determinable which admits in some way of being defined, and only that admits of being defined which admits of being bounded, and the bounded is a finite — it must

^{*)}See Carnot: *Reflexions sur la Méthaphysique du Calcul Infinitésimal*. Paris 1797 (übersetzt von Hauf. Frankf. a. M. 1800). Versuch einer Philosophie der Mathematik — mit einer Kritik der Aufstellungen Hegel’s über den Zweck und die Natur der höheren Analysis — von H. Schwartz. Halle 1853.

constantly appear to the understanding as though only the finite were conceivable, while the infinite, the unbounded, were really not to be conceived at all, but, like the ancients' hyle (τὸ ἄπειρον), to be known by a kind of spurious thinking (λογισμὸς νόθος).[†] Where this misunderstanding has taken hold there can of course be no question of entering more closely into the problem of infinity; to see the problem's philosophical significance one must absolutely have at least an inkling of the dialectical reciprocity of the finite and the infinite. "An angle of 90^0 can be divided into infinitely many infinitely small angles, and so can an angle of 1^0 ; a square mile can be divided into infinitely many infinitely small squares, and so can a square inch: what follows from this? If the emphasis is laid on the resolution of the finite into the infinite, the consequence does indeed seem to be that there is no longer any difference between the angle of 90^0 and the angle of 1^0 , that the part becomes equal to the whole — that the boundary, in other words, is effaced. But if the emphasis is laid instead on the infinite's existing in and through the finite, then there is not only a difference between finite and finite, between the part and the whole, but at the same stroke a definite difference shows itself between infinite and infinite, with respect to the infinitely small as well as to the infinitely great. The one infinite number of infinitely small angles may then be as different from the other as 1^0 from 90^0 ; the one infinite number of infinitely small squares as different from the other as a square inch from a square mile. That the boundary is sublated in the finite means that it is posited in the infinite; that the infinite takes the boundary up into itself means that the finite is posited. The dialectical unity shows itself in this, that the boundary belongs as much to the infinite as to the finite" (Mathm. og Dial. p. 33). The fundamental thought that the infinite not only effaces the boundaries of finitude but also takes the sublated boundary-determinations up as articulating moments in its own development of concepts is, from the standpoint of the operations, sufficiently designated by saying that the infinite is analysed.

b). The Analysis of the Infinite.

[†] *Translator's note.* The phrase is Plato's, *Timaeus* 52b, where the receptacle of becoming is said to be apprehended by a "bastard reasoning" — λογισμῷ τινι νόθῳ — and hardly to be believed in. Nielsen pairs it with *hyle*, matter, and with the Aristotelian τὸ ἄπειρον, the unlimited.

B.**Philosophy's Relation to Chemistry:
The Problem of Change.**

a) Chemistry as a Distinct Science.

α) The Origin of Chemistry.

β) Chemical Forces.

γ. Chemical Combinations.

b) The Problem of Change in Chemistry.

α) Atom and Equivalent.

β) Atomic Constitution and Crystal Form.

γ) Atom and Substance.

c) The Problem of Change in Philosophy.

α) The Atom as Point of Discretion.

β) The Atom as Point of Function.

γ) The Atom as the Substantial Moment.

Result.**C.****Philosophy's Relation to Physiology:
The Problem of Life.**

a) The Essence of Life.

α) The Organism.

β. The Organic Type.

γ) The Organic Purpose.

b) The Stages of Life.

III.

Philosophy and Academic Study.

Contents.